Document Translation Example

Layout-preserving bilingual PDF regression · FluentRead Research

ABSTRACT

A document translator should preserve columns, figures, captions, and reading order while replacing text inside the corresponding page regions.

1 PAGE-AWARE TRANSLATION

FluentRead parses the PDF text layer into layout blocks with page coordinates. Each translated block is fitted back into its original region instead of being shown as a detached transcript.

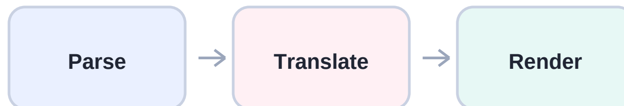
2 READING EXPERIENCE

The reader presents the untouched source page and the translated page side by side. Pagination, multi-column flow, graphics, and captions remain visible for visual comparison.

3 EXPORT CONTRACT

A bilingual download contains one wide page for every source page: the original is placed on the left and the translated layout is placed on the right.

FIGURE 1 DOCUMENT PIPELINE



Text extraction is an internal step, not the final reading interface.

4 VISUAL INVARIANTS

Regression evidence must prove that the translated page retains the figure, column boundaries, title hierarchy, and page dimensions. A generic list of OCR-like text fragments does not satisfy this contract.

Regression Coverage

Expected invariants for preview and downloadable PDF output

Capability	Expected	Evidence
Page dimensions	Preserved	Source and translated pages share the same height
Two-column flow	Preserved	Blocks remain in their original left or right column
Figures and charts	Preserved	Non-text page graphics remain in place

5 MULTI-COLUMN CONTENT

Page boundaries and paragraph order survive parsing. The translated page uses the same canvas as the source page before text regions are replaced, which leaves lines, colored shapes, and charts untouched.

6 FAILURE BOUNDARY

Image-only scans without a usable text layer are reported clearly. This beta does not pretend that extracted OCR text alone is a translated PDF document.

FIGURE 2 BLOCK RETENTION

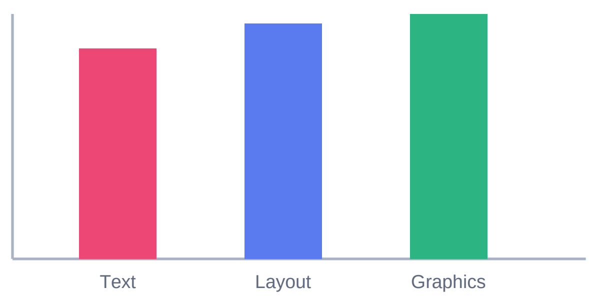


Figure 2. Layout and non-text graphics remain anchored to the page.